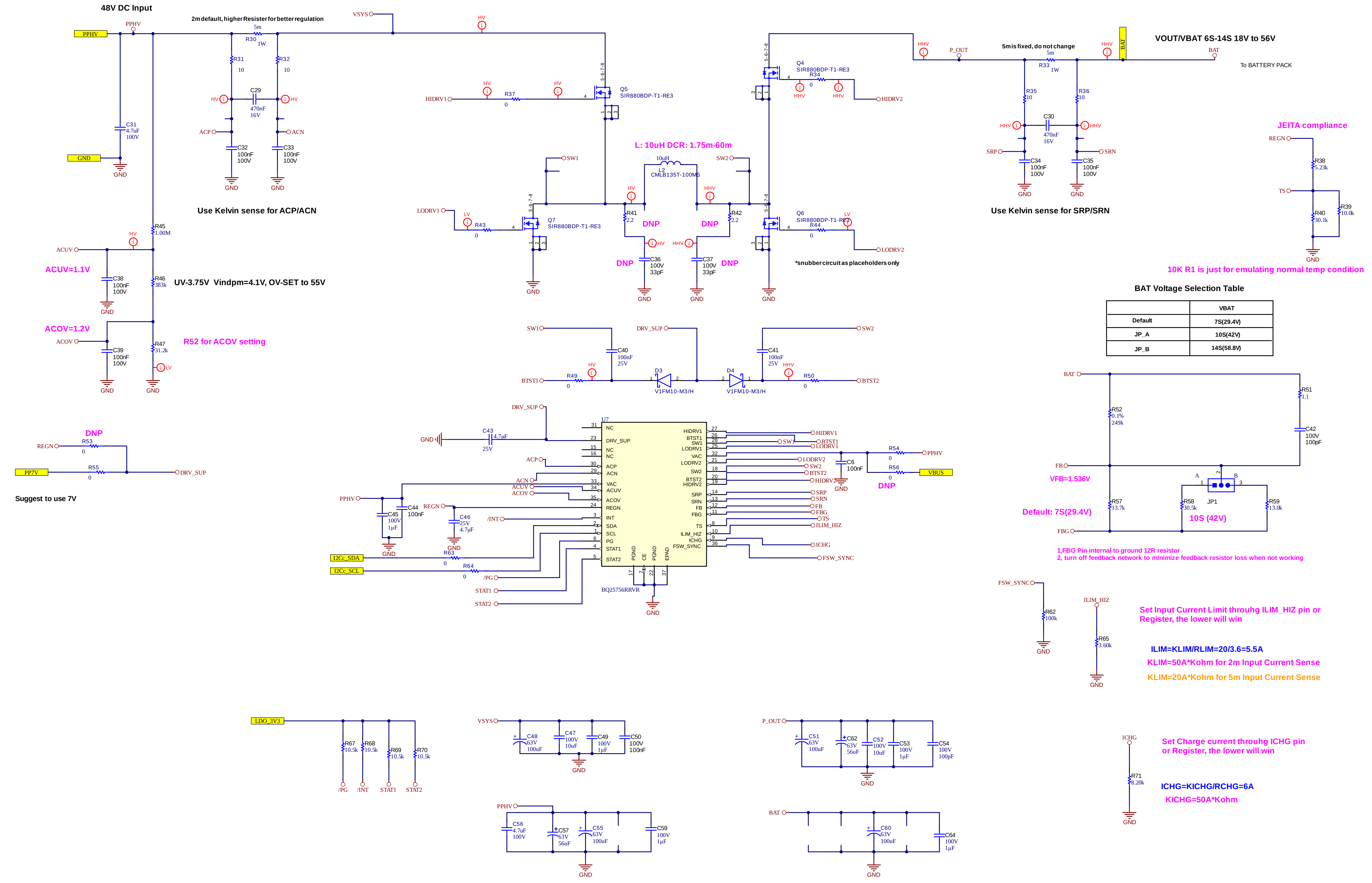

48V DC Input

2m default, higher Resister for better regulation

Use Kelvin sense for ACP/ACN

L: 10uH DCR: 1.75m-60m

DNP

*snubber circuit as placeholders only

5m is fixed, do not change

VOUT/VBAT 6S-14S 18V to 56V

To BATTERY PACK

Use Kelvin sense for SRP/SRN

JEITA compliance

10K R1 is just for emulating normal temp condition

ACUV=1.1V

UV-3.75V Vindpm=4.1V, OV-SET to 55V

ACOV=1.2V

R52 for ACOV setting

Suggest to use 7V

BAT Voltage Selection Table

	VBAT
Default	7S(29.4V)
JP_A	10S(42V)
JP_B	14S(58.8V)

VFB=1.536V

Default: 7S(29.4V)

10S (42V)

1,FBG Pin internal to ground 12R resistor
2, turn off feedback network to minimize feedback resistor loss when not working

Set Input Current Limit throuhg ILIM_HIZ pin or Register, the lower will win

ILIM=KLIM/RLIM=20/3.6=5.5A

KLIM=50A*Kohm for 2m Input Current Sense

KLIM=20A*Kohm for 5m Input Current Sense

Set Charge current throuhg ICHG pin or Register, the lower will win

ICHG=KICHG/RCHG=6A

KICHG=50A*Kohm

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Orderable:	Designed for: Public Release	Mod. Date: 15/08/2026	TEXAS INSTRUMENTS
TID #: N/A	Project Title: 240W USB PD3.1 DCDC Charger		
Number: PMP41115 Rev:	Sheet Title:		
SVN Rev: Not in version control	Assembly Variant: [No Variations]	Sheet: 3 of 4	
Drawn By:	File: 3_BQ25756 Charger.SchDoc	Size: A2	http://www.ti.com
Engineer: Max Wang	Contact: http://www.ti.com/support		© Texas Instruments 2024

Name	Description	Designator	Footprint	LibRef	Quantity
C1005X5R1H104K050BB	CAP, CERM, 0.1 uF, 50 V, +/- 10%, X5R, 0402	C1, C15	0402L	CMP-0006279-2	2
GRM188R72A104KA35J	CAP, CERM, 0.1 uF, 100 V, +/- 10%, X7R, 0603	C2, C11	0603	CMP-0007470-4	2
C2012X5R2A475K125AC	4.7 µF ±10% 100V Ceramic Capacitor X5R 0805 (2012 Metric)	C3, C80	FP-C2012X5R2A475K125AC_0805-MFG	CMP-0098058-1	2
C0603X103K1RACTU	CAP, CERM, 0.01 uF, 100 V, +/- 10%, X7R, 0603	C4, C5	0603L	CMP-0006977-4	2
HMK107B7104KAHT	CAP, CERM, 0.1 µF, 100 V,+/- 10%, X7R, AEC-Q200 Grade 1, 0603	C6, C32, C33, C34, C35, C38, C39, C44	0603_HV	CMP-0081011-1	8
CGA1A2X7R1H221K030BA	CAP, CERM, 220 pF, 50 V, +/- 10%, X7R, AEC-Q200 Grade 1, 0201	C8, C9	0201_033L	CMP-0009652-3	2
885012205037	CAP, CERM, 0.1 uF, 16 V, +/- 10%, X7R, 0402	C10	0402	CMP-0006228-1	1
CGA2B3X7R1H104K050BB	CAP, CERM, 0.1 uF, 50 V, +/- 10%, X7R, AEC-Q200 Grade 1, 0402	C12	0402L	CMP-0006325-2	1
GRM188R61E106MA73D	CAP, CERM, 10 uF, 25 V, +/- 20%, X5R, 0603	C13, C14	0603L	CMP-0007344-5	2
CL05A106MP5NUNC	CAP, CERM, 10 uF, 10 V, +/- 20%, X5R, 0402	C16, C17, C18	0402_070	CMP-0006339-4	3
C2012X7S2A105K125AB	CAP, CERM, 1 uF, 100 V, +/- 10%, X7S, 0805	C19, C20, C28	0805_HV	CMP-0007891-2	3
C3216X7S2A225K160AB	CAP, CERM, 2.2 uF, 100 V, +/- 10%, X7S, 1206	C21	1206_180	CMP-0008724-2	1
GRM155R62A104KE14D	CAP, CERM, 0.1 uF, 100 V, +/- 10%, X5R, 0402	C22	0402	CMP-0006618-1	1
0402YD105KAT2A	CAPACITOR, X5R, 0402, 16V, 1UF	C23	FP-0402YD105KAT2A_0402-IPC_B	CMP-00860000-2	1
C0402C100C8GACTU	CAP, CERM, 10 pF, 10 V,+/- 2.5%, C0G/NP0, 0402	C24	0402	CMP-0064914-1	1
CL10A226MP8NUNE	CAP, CERM, 22 uF, 10 V, +/- 20%, X5R, 0603	C25, C26	0603	CMP-0007108-2	2
CGA3E3X5R1A225K080AB	CAP, CERM, 2.2 uF, 10 V, +/- 10%, X5R, AEC-Q200 Grade 3, 0603	C27	0603	CMP-0007091-5	1
C0603C474K4RACTU	CAP, CERM, 0.47 uF, 16 V, +/- 10%, X7R, 0603	C29, C30	0603	CMP-0006944-4	2
GCJ32DC72A475KE01L	4.7uF ±10% 100V Ceramic Capacitor X7S 1210 (3225 Metric)	C31, C56	FP-GCJ32DC72A475KE01L_1210-MFG	CMP-0093675-1	2
06031A330JAT2A	CAP, CERM, 33 pF, 100 V, +/- 5%, C0G/NP0, 0603	C36, C37	0603_HV	CMP-0006722-4	2
06033C104KAT2A	CAP, CERM, 0.1 uF, 25 V, +/- 10%, X7R, 0603	C40, C41	0603_HV	CMP-0006753-4	2
GCM1885C2A101JA16D	CAP, CERM, 100 pF, 100 V, +/- 5%, C0G/NP0, AEC-Q200 Grade 1, 0603	C42	0603_HV	CMP-0007130-4	1
CGA4J1X7R1E475K125AE	Cap Ceramic 4.7uF 25V X7R 10% Pad SMD 0805 +125°C Automotive T/R	C43, C46	FP-CGA4J1X7R1E475K125AE_0805-MFG	CMP-0093626-1	2
08051C105K4Z2A	CAP, CERM, 1 µF, 100 V,+/- 10%, X7R, AEC-Q200 Grade 1, 0805	C45, C49, C53, C59, C64	0805_150	CMP-0861070-1	5
C3225X7R2A106K250AC	10 µF ±10% 100V Ceramic Capacitor X7R 1210 (3225 Metric)	C47, C52	FP-C3225X7R2A106K250AC_1210-MFG	CMP-0096021-1	2
PCR1J101MCL1GS	100uF 63V Polymer ALUM Electrolytic Capacitor, CR Series 4000h 10.3x10.3x12.6mm	C48, C51, C55, C60	FP-PCR1J101MCL1GS_SMD_RADIAL-MFG	CMP-0089206-1	4
HMK107B7104KAHT	CAP, CERM, 0.1 µF, 100 V,+/- 10%, X7R, AEC-Q200 Grade 1, 0603	C50	0603L	CMP-0081011-1	1
CGA3E2C0G2A101J080AA	Multilayer Ceramic Capacitors MLCC - SMD/SMT CGA 0603 100V 100pF C0G 5% AEC-Q200	C54	0603	CMP-0007237-3	1
EEHZC1J560P	CAP, Polymer Hybrid, 56 uF, 63 V, +/- 20%, 30 ohm, 10x10 SMD	C57, C62	SM_RADIAL_10BMM	CMP-0012603-3	2
0.1u	CAP, CERM, 0.1 uF, 100 V, +/- 10%, X7R, 0603	C65, C67	G-188_MUR	GRM188R72A104KA35J	2
1u	1 ÂµF Â±10% 100V Ceramic Capacitor X7S 0805 (2012 Metric)	C66	CAPC2012X145N	C2012X7S2A105K125AB	1
V1FM10-M3/H	Diode Schottky 1A Surface Mount DO-219AB (SMF)	D3, D4	FP-V1FM10-M3-H_DO-219AB-MFG	CMP-0093639-2	2
DX07S024JJ2R1300	Receptacle, USB 3.1 Type C, R/A, Gold, SMT	J1	JAE_DX07S024JJ2R1300	CMP-0085712-2	1
SSW-106-01-T-S	Receptacle, 6x1, 2.54mm, Tin, TH	J2	Samtec_SSW-106-01-x-S	CMP-0075621-1	1
PEC05DAAN	Header, 100mil, 5x2, Tin, TH	J3	CONN_PEC05DAAN	CMP-0054550-2	1
BM04B-SRSS-TB	Description	J5	CONN4_BM04B-SRSS_JST	BM04B-SRSS-TB	1
B4B-PH-K-S		J6	JST_B4B-PH-K-S	B4B-PH-K-S	1
PEC03SAAN	Header, 100mil, 3x1, Tin, TH	JP1	CONN_PEC03SAAN	CMP-0002338-1	1
742792021	Ferrite Bead, 22 ohm @ 100 MHz, 6 A, 0805	L1	0805L	CMP-0003072-1	1
CMLB135T-100MS	Power Choke Coil 10uH 20% 9A 22mOhm	L2	FP-CMLB135T-100MS_SMT2_13MM45_12Mm6-MFG	CMP-0096007-1	1
CSD18543Q3A	MOSFET, N-CH, 60 V, 35 A, DNH0008A (VSONP-8)	Q1	DNH0008A	CMP-0000826-2	1
2N7002DW	MOSFET, 2-CH, N-CH, 60 V, 0.115 A, SOT-363	Q2, Q3, Q9	SOT-363	CMP-0000770-1	3
SIR880BDP-T1-RE3	N-Channel 80 V 18.6A (Ta), 70.6A (Tc) 5W (Ta), 71.4W (Tc) Surface Mount PowerPAK® SO-8	Q4, Q5, Q6, Q7	FP-SIR880BDP-T1-RE3_SO-8-MFG	CMP-0096839-1	4
ERJ-2RKF1001X	RES, 1.00 k, 1%, 0.1 W, 0402	R1, R2	0402	CMP-0026657-1	2
RC0402FR-072K2L	RES, 2.20 k, 1%, 0.063 W, 0402	R3, R4, R6, R7	0402	CMP-0026729-1	4
CRCW040210K0DHEDP	RES, 10.0 k, 0.5%, 0.063 W, 0402	R5, R8	0402	CMP-0026769-1	2
CRCW02010000Z0ED	RES, 0, 5%, 0.05 W, 0201	R10, R11	0201L	CMP-0027008-4	2
RC0201FS-7D100KL	RES, 100 k, 1%, 0.05 W, 0201	R12	0201L	CMP-0028069-1	1
RT0603BRD07470RL	RES, 470, 0.1%, 0.1 W, 0603	R13, R16	0603	CMP-0024263-1	2
RC0603FR-07100KL	RES, 100 k, 1%, 0.1 W, 0603	R14, R15	0603L	CMP-0022735-5	2
RC0402JR-070RL	RES, 0, 5%, 0.063 W, 0402	R18, R20, R48, R60, R66, R98, R99, R100, R103, R104, R107, R109, R110	0402	CMP-0027004-1	13
RC0603JR-070RL	RES, 0, 5%, 0.1 W, 0603	R21, R25	0603	CMP-0025885-5	2
CMP-0027004-2	RES, 0, 5%, 0.063 W, 0402	R22, R23, R26, R27	0402	CMP-0027004-2	4
RT0603BRD0760K4L	RES, 60.4 k, 0.1%, 0.1 W, 0603	R24	0603	CMP-0024382-2	1
CRCW040219K6FKED	RES, 19.6 k, 1%, 0.063 W, 0402	R28	0402	CMP-0026068-2	1
ERJ-2RKF1002X	RES, 10.0 k, 1%, 0.1 W, 0402	R29	0402	CMP-0026658-1	1
WSL25125L000FEA	Res Metal Strip 2512 0.005 Ohm 1% 1W 110ppm/C Molded SMD SMD Embossed Plastic T/R	R30, R33	FP-WSL25125L000FEA_2512-MFG	CMP-0096841-1	2
CRCW060310R0FKEB	RES Thick Film, 10Ω, 1%, 0.1W, 100ppm/°C, 0603	R31, R32, R35, R36	FP-CRCW060310R0FKEB_0603-MFG	CMP-0093685-1	4
CRCW06030000Z0EA	Thick Film Resistors - SMD 1/10watt ZEROohm Jumper	R34, R37, R43, R44, R49, R50, R54, R55	FP-CRCW06031R50FKEAHP_0603-MFG	CMP-0093629-1	8
RC0603FR-075K23L	RES, 5.23 k, 1%, 0.1 W, 0603	R38	0603	CMP-0023199-5	1
ERJ-3EKF1002V	RES, 10.0 k, 1%, 0.1 W, 0603	R39	0603	CMP-0022684-2	1
RC0603FR-0730K1L	RES, 30.1 k, 1%, 0.1 W, 0603	R40	0603	CMP-0023018-5	1
CRCW08051M00FKEAC	Thick Film Resistors - SMD 1/8Watt 1Mohms 1% Commercial Use	R45	0805_HV	CMP-0021522-3	1
RT0603BRD07383KL	RES, 383 k, 0.1%, 0.1 W, 0603	R46	0603	CMP-0024170-1	1
RT0603BRD0731K2L	RES, 31.2 k, 0.1%, 0.1 W, 0603	R47	0603	CMP-0024106-3	1
CRCW06031R10JNEA	RES, 1.1, 5%, 0.1 W, 0603	R51	0603	CMP-0025744-3	1
ERJ-PB6B2493V	Thick Film Resistors - SMD 0805 Anti-Surge Res. 0.1%, 249Kohm	R52	0805_HV	CMP-0018933-2	1
RT0603BRD0713K7L	RES, 13.7 k, 0.1%, 0.1 W, 0603	R57	0603	CMP-0023733-2	1
RT0603BRD0730K5L	RES, 30.5 k, 0.1%, 0.1 W, 0603	R58	0603	CMP-0024097-2	1
RT0603BRD0713KL	RES, 13.0 k, 0.1%, 0.1 W, 0603	R59	0603	CMP-0023735-1	1
RC0603FR-07100KL	RES, 100 k, 1%, 0.1 W, 0603	R62	0603	CMP-0022735-5	1
CRCW08050000Z0EA	Thick Film Resistors - SMD 1/8watt ZEROohm Jumper	R63, R64	FP-CRCW080510R0JNEAHP_0805-MFG	CMP-0092857-1	2
RC0603FR-073K6L	RES, 3.60 k, 1%, 0.1 W, 0603	R65	0603	CMP-0023086-5	1
RC0402FR-0710K5L	RES, 10.5 k, 1%, 0.063 W, 0402	R67, R68, R69, R70	0402	CMP-0078781-1	4
RG1608P-822-B-T5	RES, 8.20 k, 0.1%, 0.1 W, 0603	R71	0603	CMP-0023607-3	1
RC1005F104CS	RES, 100 k, 1%, 0.063 W, 0402	R97, R108	0402L	CMP-0026745-1	2
RG1608P-2493-B-T5	RES, 249 k, 0.1%, 0.1 W, 0603	R101, R102	0603	CMP-0023490-1	2
ERJ-2RKF1004X	RES, 1.00 M, 1%, 0.1 W, 0402	R105, R106	0402L	CMP-0026660-1	2
10k	D10/CRCW0402-P 50 10K 0.5% ET7 e3	R111	RC0402N_VIS	CRCW040210K0DHEDP	1
TPS26750RSMR	USB Type-C® and USB PD Controller with Integrated Power Switches Optimized for Power Applications	U1	RSM0032B-MFG	CMP-0121143-2	1
TPD4S480RUKR	USB Type-C® 48V EPR Port Protector: Short-to-VBUS Overvoltage and IEC ESD Protection	U2	RUK0020B-MFG	CMP-0034161-1	1
CAT24C512WI-GT3	EEPROM Memory IC 512Kb (64K x 8) I²C 1MHz 900ns 8-SOIC	U3	FP-CAT24C512WI-GT3_SOIC8-IPC_C	CMP-0091847-1	1
TPS7A2450DBVR	Linear Voltage Regulator IC 1 Output 200mA SOT-23-5	U4	DBV0005A-MFG	CMP-0096923-1	1
TPSM365R6RDNR	3 V to 65 V Input, 1 V to 13 V Output, 600-mA Synchronous Buck Converter Power Module in a HotRod™ QFN Package	U5	RDN0011A-MFG	CMP-0096774-1	1
TPS76333DBVR	Single Output LDO, 150 mA, Fixed 3.3 V Output, 2.7 to 10 V Input, with Low IQ, 5-pin SOT-23 (DBV), -40 to 125 degC, Green (RoHS & no Sb/Br)	U6	DBV0005A_N	CMP-0035539-1	1
BQ25756RRVR	Stand-alone or I&sup2;C-controlled 70-V bidirectional buck-boost charge controller with MPPT 36-VQFN -40 to 125	U7	VQFN36_RRV_TEX	BQ25756RRVR	1
EMC2101-R-ACZL-TR	Description	U8	MSOP8_MC_MCH	EMC2101-R-ACZL-TR	1